



BALLARI INSTITUTE OF TECHNOLOGY AND MANAGEMENT, BALLARI
All Departments



DATA SECURITY AND PRIVACY (22CS2744)

ASSIGNMENT-1

Module - I

1. Define Cryptography and explain symmetric cipher model.
OR
Draw the simplified model of symmetric encryption and explain it.
2. Define following terms
 - a) Cryptography
 - b) Ciphertext
 - c) Encryption
 - d) Decryption
 - e) Cryptanalysis
3. List and explain the 3 independent dimensions of cryptographic system.
4. List and briefly define types of cryptanalytic attacks based on what is known to the attackers.
5. Explain Caesar cipher with example.
6. Find the Ciphertext for plaintext "MEET ME AFTER THE TOGA PARTY" with key = 3, using Caesar cipher algorithm, assign a numerical equivalent to each letter.
7. Explain Monoalphabetic cipher with example.
8. Explain Playfair cipher algorithm. Find the ciphertext for plaintext "INSTRUMENTS" with key "MONARCHY".
9. Encrypt the plaintext "CRYPTOGRAPHY" using Hill Cipher algorithm with key $k = \begin{bmatrix} 9 & 4 \\ 5 & 7 \end{bmatrix}$ and decrypt the same.
10. Explain Polyalphabetic and One Time Pad cipher with example.
11. Distinguish between.
 - a) Confusion and Diffusion.
 - b) Block cipher and Stream cipher.
12. Explain Feistel encryption and decryption algorithm, with neat diagram.
13. Explain with neat diagram DES encryption and decryption algorithm.

Module – II

1. With a neat diagram, explain five ingredients of a public key cryptography.
2. What requirements must a public key cryptosystems fulfill to be a secure algorithm?
3. Differentiate Public key encryption and Conventional encryption. Classify and explain public key cryptosystems.
4. With relevant diagram, explain Authentication and secrecy in public key crypto systems.
5. Explain the requirements and applications of public key cryptography.
6. Explain RSA Algorithm operation in detail. Perform an encryption of plain text and decryption of cipher text using RSA algorithm for $p=3$, $q=11$, $e=7$ and $M=5$.
7. Explain Diffie-Hellman key exchange algorithm. Apply Diffie- Hellman key exchange algorithm for $q=71$, its primitive root $\alpha=7$. A's private key is 5, B's private key is 12. Find i) A's Public key ii) B's Public key iii) Shared secret key
8. "Diffie-Hellman key exchange is vulnerable to Man-in-the-Middle attack". Substantiate it with sequence diagram.

(OR)

Explain Man-in-the-Middle attack with example.

9. Explain Elgamal cryptosystem.
10. Discuss elliptic curve cryptography for analog of Diffie-Hellman key exchange and explain with neat steps.
11. Explain elliptic curve over real numbers.
12. Explain pseudorandom number generation based on asymmetric cipher.

Module – III

1. With relevant diagram, explain key distribution scenario.
2. Explain Symmetric key distribution using Symmetric encryption.
3. Explain issues with KDC (Key Distribution Center)

(OR)

Explain:

- i) Hierarchical key control
- ii) Session key lifetime
- iii) A transparent key control scheme
- iv) Decentralized key control

NOTE: Any of these or similar to these can be asked in the exam